Other Related Tests

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- A natural test statistic is based the asymptotic distribution of $\hat{\theta}$. Consider the statistics

$$\chi_W^2 = \{\sqrt{nI(\hat{\theta})}\,(\hat{\theta} - \theta_0)\}^2.$$

Under H_0, χ_W^2 has an asymptotic $\chi^2(1)$. The decision rule is

$$\text{Reject } H_0 \text{ in favor of } H_1 \text{ if } \chi_W^2 \geq \chi_\alpha^2(1)$$

This test is often referred as a **Wald**-type test, after Abraham Wald, who was a prominent statistician of the 20th century.

- The third test is called **score**-type test, which is often referred to as Rao's score test, after another prominent statistician C.R.Rao. In our notation,

$$\frac{1}{\sqrt{n}} l'(\theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \frac{\partial \log f(X_i; \theta)}{\partial \theta}.$$

Define the statistics

$$\chi_R^2 = \left(\frac{l'(\theta_0)}{\sqrt{nI(\theta_0)}}\right)^2.$$

Under H_0, χ_R^2 has an asymptotic $\chi^2(1)$. The decision rule is

$$\text{Reject } H_0 \text{ in favor of } H_1 \text{ if } \chi_R^2 \geq \chi_\alpha^2(1)\ .$$

Examples

- Let $X_1, \ldots, X_n$ be a random sample from Beta$(\theta, 1)$ with pdf $f(x;\theta) = \theta x^{\theta-1}$, $0 < x < 1$. We will use this pdf to illustrate three test statistics for the hypotheses

$$H_0 : \theta = 1 \qquad vs. \qquad H_1 : \theta \neq 1.$$

Recall mle $\hat{\theta} = -n/\sum_{i=1}^{n} \log X_i$. The value of the likelihood function at the mle is

$$L(\hat{\theta}) = (-\sum_{i=1}^{n} \log X_i)^{-n} \exp\{-\sum_{i=1}^{n} \log X_i\} \exp\{n(\log n - 1)\}.$$

Also, $L(1) = 1$. The likelihood ratio test statistic $\Lambda = 1/L(\hat{\theta})$ so that

$$\chi_L^2 = -2\log\Lambda = 2\{-\sum_{i=1}^{n}\log X_i - n\log(-\sum_{i=1}^{n}\log X_i) - n + n\log n\}.$$

Examples

Recall that $I(\theta) = \theta^{-2}$. For wald-type test,

$$\chi_W^2 = (\sqrt{\frac{n}{\hat{\theta}^2}}(\hat{\theta} - 1))^2 = n\{1 - \frac{1}{\hat{\theta}}\}^2.$$

Examples

Finally, recall that

$$l'(1) = \sum_{i=1}^{n} \log X_i + n,$$

Hence the scores type test statistics is

$$\chi_R^2 = \{\frac{\sum_{i=1}^{n} \log X_i + n}{\sqrt{n}}\}^2.$$

Examples

- Likelihood tests for the Laplace location model: Consider the location model

$$X_i = \theta + \epsilon_i, i = 1, \ldots, n,$$

where ϵ_is are iid with Laplace pdf $f(z) = 2^{-1}\exp\{-|z|\}$. Consider the test

$$H_0 : \theta = \theta_0 \qquad vs \qquad H_1 : \theta \neq \theta_0.$$

Recall the mle is $Q_2 = med\{X_1, \ldots, X_n\}$. It follows that

$$L(Q_2) = 2^{-n}\exp\{-\sum_{i=1}^{n}|X_i - Q_2|\}.$$

So,

$$-2\log\Lambda = 2[\sum_{i=1}^{n}|X_i - \theta_0| - \sum_{i=1}^{n}|X_i - Q_2|]$$

Examples

The fisher information is $I(\theta) = 1$. Thus

$$\chi_W^2 = [\sqrt{n}(Q_2 - \theta_0)]^2.$$

For the score test, we have

$$\frac{\partial \log f(X_i - \theta)}{\partial \theta} = sgn(X_i - \theta).$$

$$\chi_R^2 = (S^*)^2/n = \frac{1}{n}\sum_{i=1}^{n} sgn(X_i - \theta_0).$$